ROADMAP

Helix OTA — Emulation Test-Fabric — Phased Implementation Roadmap

Field	Value
Revision	1
Last modified	2026-06-10T14:10:00Z
Status	active — plan (PWUs, not yet implemented)
Status summary	Phased PWU roadmap (§11.4.58) building the fabric in `DESIGN.md`. Each phase lists scope, the `containers`-submodule extension, the test types + captured-evidence plan, the independent-review gate (§11.4.125/ §11.4.142), and its host-gating (dev-host-now / Linux-KVM / real-hardware). NO phase is “done” without real captured evidence (§11.4).
Related	DESIGN.md; TEST_COVERAGE_PLAN.md; SCHEMA.sql; REPORT.md

Sequencing principle

Land the **dev-host-runnable** tiers first (immediate value, no host gate), then the Linux-KVM CI tiers (highest fidelity), then real-hardware HIL. Each phase is a self-contained PWU with its own RED→GREEN evidence and a structurally-separate review before merge.

Phase	Scope (deliverable)	containers-submodule extension	Test types + evidence (anti-bluff)	H
P0 — fabric floor (DONE)	T0 `ota-device-emulator` + full-lifecycle + fleet +	reuse `pkg/boot/compose/health`	e2e + integration + scaling; transcripts under `docs/qa/` (already shipped: full-lifecycle, fleet, recall-recovery, telemetry-fields)	**d** **n**

Phase	Scope (deliverable)	containers-submodule extension	Test types + evidence (anti-bluff)	H
	recall→recovery e2e			
P1 — T1 Android AVD (HVF) local	boot arm64-v8a AVD on HVF; drive `ota-android-agent`/bridge decision logic + ADB UI automation; resolve the `UNCONFIRMED:` GSI-A/B question (REPORT §2.2) via an empirical GSI test	extend `pkg/emulator` (arm64-v8a HVF profile + AVD-lease)	unit+integration+e2e+ui; evidence = ADB dumpsys + `update_engine_client` state + screen capture (§11.4.107 liveness where UI)	d n
P2 — Tfw firmware/ U-Boot	QEMU `-machine virt`+edk2 boot of U-Boot slot-switch logic; Renode peripheral determinism (future Linux target)	reuse `pkg/vm`; add Renode profile	integration+e2e; evidence = boot log + slot-state assertion	d n
P3 — `pkg/fabric` target registry + scheduler façade	exclusive target leasing (§11.4.119) + `SCHEMA.sql` registry on the pgx seam; Nomad/K8s backend adapter	**new `pkg/fabric`** (PR upstream)	unit+integration (real Postgres parity) + chaos (lease contention) + stress (N concurrent leases)	d (L (
P4 — T2 Cuttlefish (Linux-KVM CI)	real `update_engine` A/B + AVB/dm-verity + auto-rollback-on-corrupt-slot via `cvd`	**new `pkg/cuttlefish`** (PR upstream; KVM-presence gate → SKIP on non-KVM host)	e2e+full-automation+chaos (corrupt-slot→auto-rollback); evidence = real A/B slot + dm-verity + rollback trace	L C g §
P5 — Tcp control-plane isolation	Firecracker/Kata microVM pods for control-plane scaling/	reuse/extend container runtime profiles	scaling+ddos+chaos+performance+benchmarking; evidence = measured percentiles + categorised faults (§11.4.85)	L C n

Phase	Scope (deliverable)	containers-submodule extension	Test types + evidence (anti-bluff)	H
	DDoS/chaos under real isolation; gVisor server sandbox			
P6 — distributed control plane (LAVA)	one LAVA job schema driving virtual (T0/T1/T2) + physical (T3) DUTs; CI front door on self-hosted runners	**new `pkg/lava`** (PR upstream)	e2e+full-automation across the fan-out; evidence = LAVA job artifacts + per-DUT transcripts	L n w
P7 — T3 real RK3588 HIL	real board(s) behind LAVA; `redroid-rk3588` adjacent SoC app/GPU; flashes governed by §11.4.133	`pkg/lava` board dispatchers	on-device 4-phase cycle + auto-rollback on genuinely-corrupt slot; evidence per §11.4.107/§11.4.69	**r h**

Coverage & HelixQA wiring (every phase)

Each phase: (a) maps its features onto every applicable test type per `TEST_COVERAGE_PLAN.md`; (b) adds a HelixQA bank challenge dispatching to the phase's real test + scoring on captured evidence (the `tools/helixqa/run_bank.sh` runner + the canonical engine's evidence ledger); (c) registers a permanent §11.4.135 regression guard for any defect found; (d) produces docs/manual/guides/diagrams (Mermaid) kept in sync (§11.4.12/§11.4.65); (e) updates the §11.4.25 coverage ledger with MEASURED coverage (never claimed).

Honest status

Only **P0 is shipped**. P1–P7 are planned PWUs. "100% coverage of all features/flows/edge-cases" is the *target* the per-phase coverage ledger ratchets toward (§11.4.50) and is **measured, never asserted** — no phase is marked complete without real captured evidence and an independent review.